Note

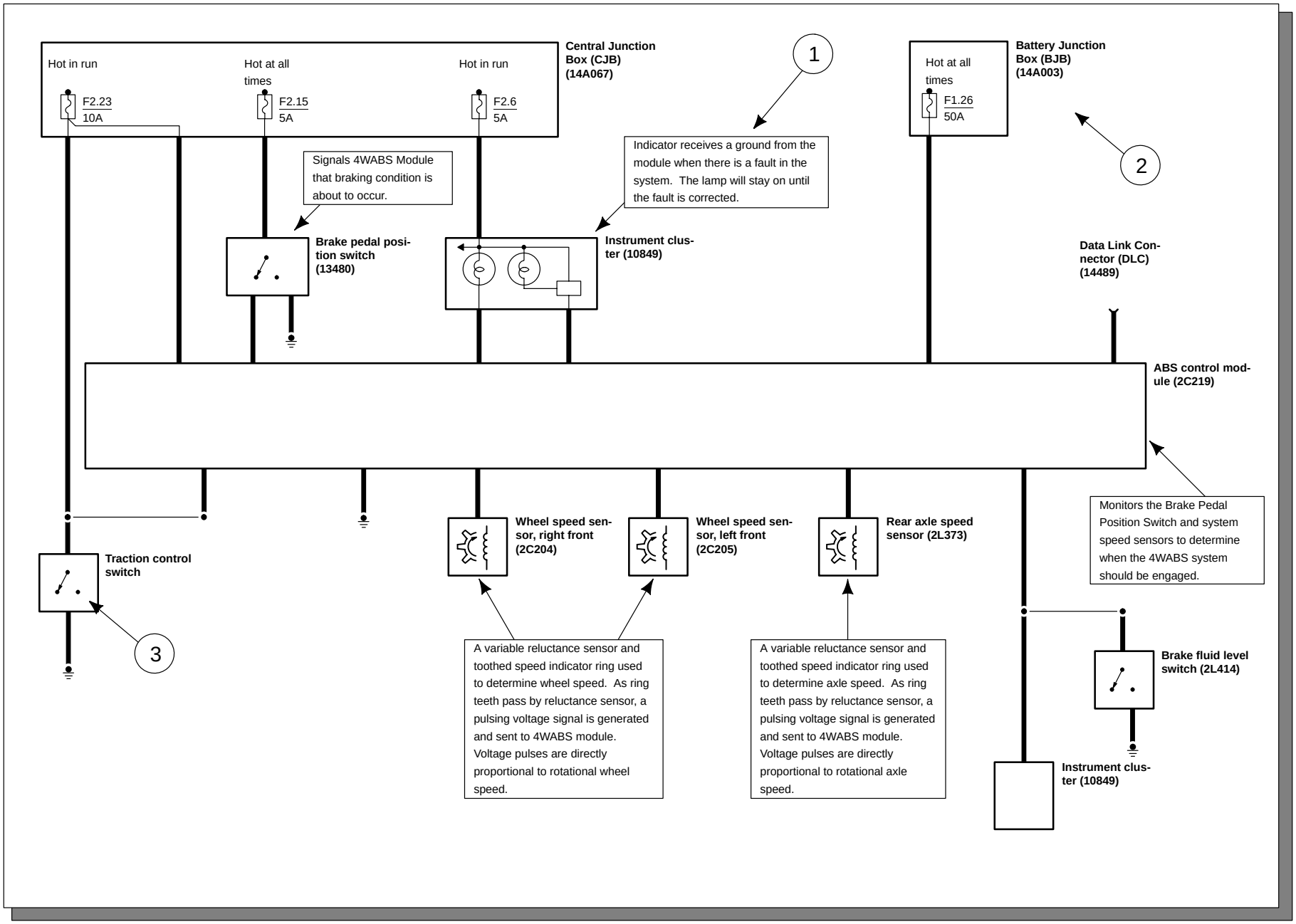
All wiring connections between components are shown exactly as they exist in the vehicles. It is important to realize, however, that no attempt has been made on the schematic to represent components and wiring as they physically appear on the vehicle. For example, a 4-foot length of wire is treated no differently in a schematic from one which is only a few inches long. Furthermore, to aid in understanding electrical (electronic) operation, wiring inside complicated components has been simplified.

Complete Circuit Operation

Each circuit is shown completely and independently in one cell. Other components which are connected to the circuit may not be shown unless they influence the circuit operation.

System Overview

Each major vehicle system includes a complete system overview prior to each set of schematic pages. It is important to realize that this is only a high level overview of the complete system connectivity. It includes component operational information (1), component name and base part number (2), and basic component internals (3). It does not include specific circuit information, connector or pin numbers, splices or grounds. That information is found on the schematic pages.


Hot in run
F2.23
10A
Hot at all times
F2.15
5A
Hot in run
F2.6
5A
Central Junction Box (CJB) (14A067)
1
Hot at all times
F1.26
50A
Battery Junction Box (BJB) (14A003)
2
Signals 4WABS Module that braking condition is about to occur.
Indicator receives a ground from the module when there is a fault in the system. The lamp will stay on until the fault is corrected.
Brake pedal position switch (13480)
Instrument cluster (10849)
Data Link Connector (DLC) (14489)
ABS control module (2C219)
Monitors the Brake Pedal Position Switch and system speed sensors to determine when the 4WABS system should be engaged.
Traction control switch
3
Wheel speed sensor, right front (2C204)
Wheel speed sensor, left front (2C205)
Rear axle speed sensor (2L373)
A variable reluctance sensor and toothed speed indicator ring used to determine wheel speed. As ring teeth pass by reluctance sensor, a pulsing voltage signal is generated and sent to 4WABS module. Voltage pulses are directly proportional to rotational wheel speed.
A variable reluctance sensor and toothed speed indicator ring used to determine axle speed. As ring teeth pass by reluctance sensor, a pulsing voltage signal is generated and sent to 4WABS module. Voltage pulses are directly proportional to rotational axle speed.
Brake fluid level switch (2L414)
Instrument cluster (10849)

Current Flow (1)

Each cell normally starts with the component that powers the circuit, such as a fuse or the ignition switch. Current flow is shown from the power source at the top of the page to ground at the bottom of the page. In order to concentrate on the essential parts, power supply and ground connections are sometimes simplified by a dashed line in the schematics. A full representation of the power supply of a fuse or the power distribution from a fuse to various components is given in cell 13 “Power Distribution”. Full representation of the ground connections are shown in cell 10 “Grounds”.

Switch Positions (2)

Within the schematic, all switches, sensors and relays are shown “at rest” (as if the Ignition Switch were OFF).

Splices (3)

A dashed line indicates that the splice is not shown completely. A reference is given to the page where the splice appears in full. It is also listed in the Index.

Component Referencing (4)

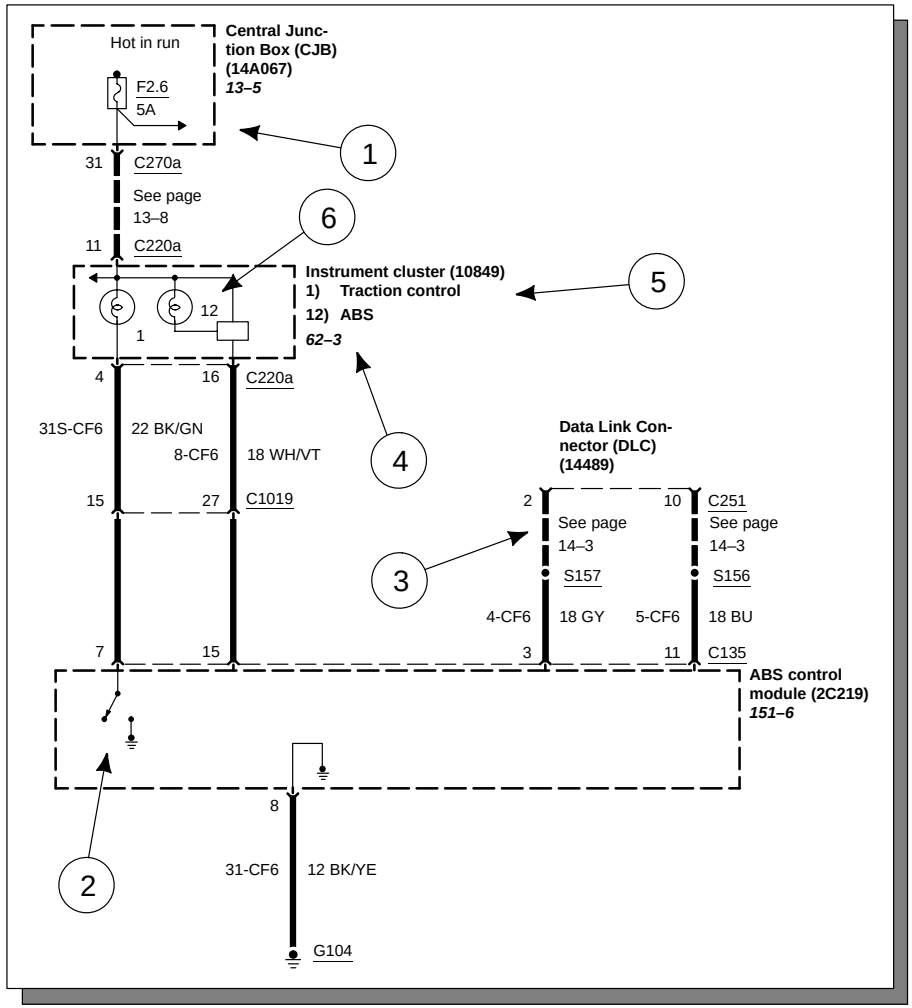
Each component on a schematic has a reference to the component location view or the page where it is shown completely. It is located to the right of each component.

Component Names, Notes and Base Part Numbers (5)

Component names are placed on the right hand side of each component. Any notes that describe switch positions or operating conditions follow the name. Descriptions of the internals of the component are also included here. The page where the component appears in full is listed in the Index. The base part number for a component is listed in parentheses next to or under a component. These part numbers will appear any place the component name appears in the publication.

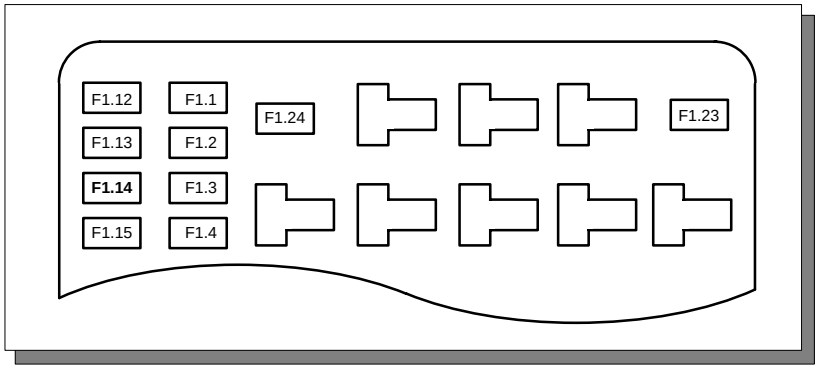
Internal Name and Function Identification Numbers (6)

Some components on each page have internal symbols with an identification number located to the right. You can identify the internal symbol or function by finding the corresponding number under the component name.



Fuse and Relay Information

Cell 11 “Fuse and Relay Information” contains a view of the fuse-/relay box in which all fuses and relays are identified.

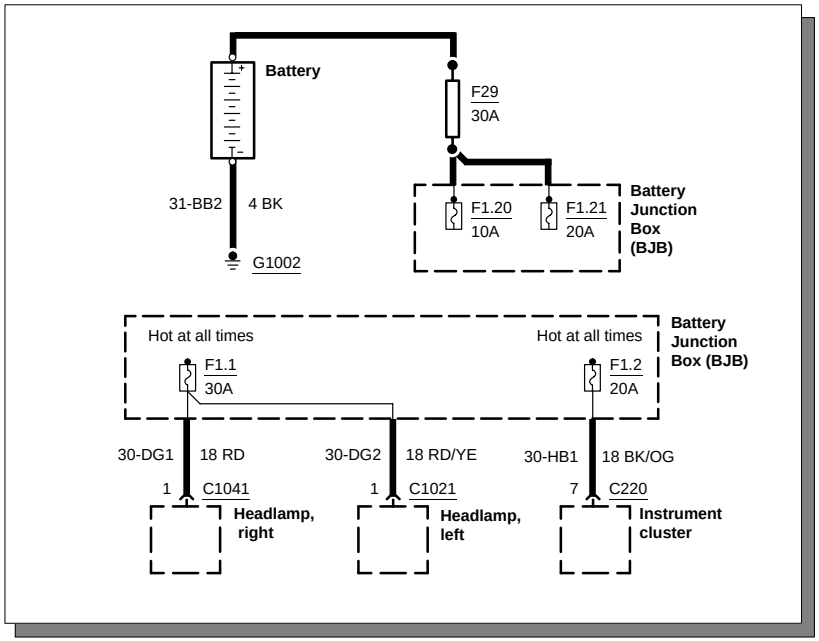


Fuse and Relay Numbering and Naming

Fuse and relay numbering and naming follow the indication of the fuse panel cover. In addition, a prefix precedes the fuse number to facilitate finding the fuse in the Component Location Charts, e.g. “F1.” precedes Battery Junction Box fuses, and “F2.” precedes Central Junction Box fuses.

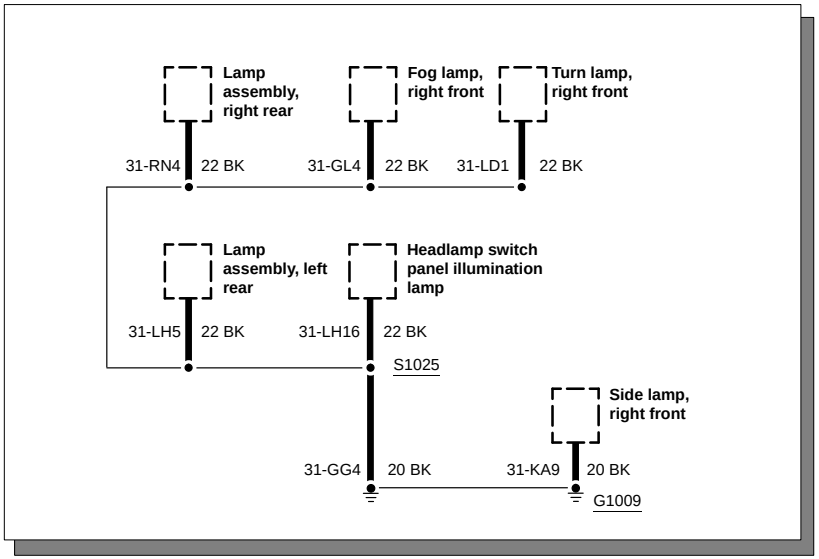
Power Distribution

Cell 13 “Power Distribution” shows the current feed circuit. The current path is shown from the battery to the ignition switch and to all fuses. It also shows the circuits protected by each fuse. The circuit is traced from the fuse to the component. All details (wires, splices, connectors) between the fuse and the first component are shown.



Ground Distribution

Cell 10 “Grounds” contains the schematics that show the complete details for each ground connection or main ground splice. This is useful in diagnosing a problem affecting several components at once (poor ground connection or ground splice). All details (wires, splices, connectors) between the ground point and the components are shown. These ground connection details are shown here in order to keep the individual cell schematics as uncluttered as possible.



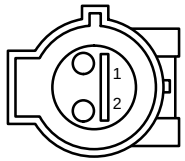
Component and Connector Information

Cell 152 “Component Location Charts” helps the user find where the various items depicted on the schematic can physically be found on the vehicle. A brief written description of the location is given, along with a reference to the component location views.
Cell 151 “Component Location Views” show the components and their connecting wires as they can be found on the vehicle.
Cell 150 “Connector Views” show the views of the pins and/or cavities of all connectors. The pin and cavity sides are shown separately as if the connector were disconnected. The color of the connector housing is indicated next to the connector number when available. The harness causal number is located above the connector view and below the connector number. The circuit function charts are located below each connector. The wiring harness designations are listed in cell 152 “Component Location Charts”.

C150

12A581

Wheel speed sensor, left front (2C205)



F02085

Pin	Circuit	Circuit Function
1	7-CF32 (YE)	Wheel speed sensor, left front (2C205) –
2	8-CF32 (WH)	Wheel speed sensor, left front (2C205) +

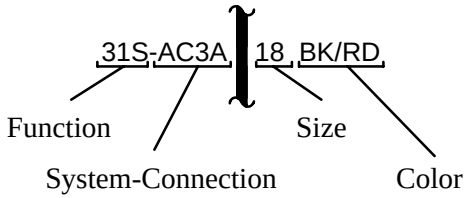
Circuit Numbering and Wire Identification

Circuit Numbering:

Ford has introduced a world-wide uniform system for circuit numbering and wire identification. The system is called Function, System-Connection, or FSC for short.

FSC was developed mainly to assist in vehicle development and production processes, but is also helpful in troubleshooting electrical circuits.

Example:



Function:

The first two digits identify the function of the wire. In this case, the function code includes the letter “S” to indicate it is an additionally switched function. The function part of the circuit number is the most helpful to the technician in troubleshooting a circuit. A complete list of function codes and their descriptions is included on the next page.

System-Connection (including branch):

Systems are related to subsets of the vehicle circuitry. A complete list of the system codes can be found on page 3-8.

Immediately following the system letter code is the connection number specific to that system. A branch identification is used to differentiate wires in one connection with the same function.

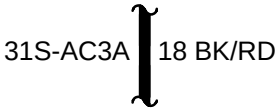
Wire Identification (wire colors):

The wire identification consists of a basic color and an identification color, and is determined directly from the wire’s circuit number. In the schematics the wire colors are indicated next to the wires.

The basic and identification colors are abbreviated using the international norm IEC 757. The abbreviations are listed in cell 4 “Symbols”.

Each function code has a specific basic color associated with it which can also be seen in the chart of function codes on the next page. The identification color is a colored stripe used to differentiate several wires with the same function within one component connector.

Using the circuit number and wire identification used in the above example we know the following information about the wire:



Function: 31 = Ground
S = Additionally switched circuit

System: AC = Headlamp leveling

Connection: 3 = Switch connection
A = Branch

Size: 18 = Gauge size

Color: BK = Basic color black
(determined by function 31)
RD = Identification color red
(red stripe)

Function Codes

Number	Basic color	Description
1	WH	Audio signal positive
2	GY	Audio signal negative
3	BN	Audio signal reference (return/shield)
4	GY	Data link, bus positive
5	BU	Data link, bus negative
6	NO	Data link, bus shield
7	YE	Voltage D.C. supplied at all times (sensor)
8	WH	Sensor signal
9	BN	Sensor signal return (reference/ground)
10	GY	Signal 2 (Y–direction, reverse, downward)
14	VT	Voltage supplied in Start and Run (overload protected)
15	GN	Voltage, ignition switch in Start or Run (not overload protected)
20	PK	Voltage, hot in run only
29	OG	Voltage supplied at all times (overload protected)
30	RD	Voltage supplied at all times (not overload protected)
31	BK	Ground
32	WH	Switched between: Battery voltage, Ground or Open
33	YE	Switched between: Battery voltage, Ground or Open
34	BU	Switched between: Battery voltage, Ground or Open
35	VT	Switched between: Battery voltage, Ground or Open
48	NO	Drain wire
49	BU	Pulsed power feed (overload protected)
50	GY	Voltage, ignition switch in Start (not overload protected)
59	GY	AC Power
63	WH	Variable D.C. voltage (overload protected)
64	BU	Variable D.C. voltage (not overload protected)
74	BU	Voltage in Run or Accessory (overload protected)
75	YE	Voltage: Ignition switch in Accessory or Run
86	OG	Trailer battery charge
87	BN	Trailer: Tail, License, Side, Clearance and I.D.
88	YE	Trailer: Left Turn and Stop if Independent Stop not provided
89	OG	Voltage supplied at all times (overload protected electronic module)
90	RD	Voltage supplied at all times (not overload protected electronic module)
91	BK	Ground
94	VT	Voltage, ignition switch in Start or Run (overload protected electronic module)
95	GN	Voltage, ignition switch in Start or Run (not overload protected electronic module)
96	GN	Trailer: Right Turn and Stop if Independent Stop not provided
97	RD	Trailer: Independent Stoplamp
98	BU	Electric trailer brakes
99	WH	Trailer ground (including for Aftermarket builder)

System Codes

	Actuated Systems (high current)
AA	Door locks and decklid (including: keyless entry/ global closing)
AB	Four–wheel drive
AC	Headlamp leveling
AD	Power mirrors
AG	Power sliding roof
AH	Power seats
AJ	Power windows
AL	Power steering column
AM	Power doors
	Basic Systems
BA	Charge (including: ammeter/voltmeter)
BB	Start
	Chassis
CC	Ride suspension
CD	Self–levelling suspension
CE	Power steering (variable automatic power assist)
CF	Traction (including: anti–lock/brakes)
CG	Brakes
	Distribution Systems
DA	Splices distribution
DB	Busbars
DC	Circuit protection – Battery Junction Box (BJB)
DD	Circuit protection – Central Junction Box (CJB)
DE	Ground
DF	Junction block
DG	Circuit protection – Auxiliary fuse box
DH	Distribution – Auxiliary fuse box
DJ	Auxiliary distribution
DK	Integrated components
	Data Bus Systems
EA	ACP Bus
EB	ALT Bus
EC	CAN Bus
ED	DCL Bus
EE	ISO Bus
EG	SCP Bus
	Climate Systems
FA	Climate 1
FB	Climate 2
FC	Climate, auxiliary heater
	Gauge Systems
GA	Gauges – level/pressure/temperature
GB	Miscellaneous (including: speedometer/clock)
GC	Indicator – level/pressure/temperature
GD	Temperature
GE	Auxiliary warning
GG	Instrument cluster
GH	Trip computer
GJ	Horn (13832)
GK	Navigation Main
GL	Anti–theft
GM	Audio warning (including: buzzers/chimes)
GN	Parking aid
GP	Vehicle emergency monitoring
	Heated Systems
HA	Auxiliary (including: air/filter/lighter)
HB	De–ice (including: ejectors/mirrors/washer jets/windows)
HC	Heated seats
	Climate 1
JA	Air bags
	Wiper/washer
KA	Wiper/washer
	Lighting
LA	Control and instrument
LB	Courtesy (glove box/hood/luggage comp./map/ recharg. flashlight/vanity)
LC	Entry (including: interior/floor/door)
LD	Fog/flood/driving
LE	Head (including: concealed/dim–dip/daytime running)
LF	Parking/marker/license/clearance
LG	Turn signal/cornering/hazard/stop/reverse
LH	Instrument illumination
LK	Switch illumination
	Entertainment
MC	Communication
MD	Entertainment
	Special Vehicles (unique)
NA	Ambulance
NB	Police
NC	Taxi
ND	Trailer
	Powertrain Control Systems
PA	Engine controls
PG	Vehicle speed and engine RPM
	Powertrain Engine Control Gas/Diesel Connections
RA	Connectors
RB	Destination notes
RD	Diodes, fuses, indicators, resistors, heaters
RE	Modules, controllers
RG	Motors
RH	Relay
RJ	Sensor
RL	Solenoids
RN	Splices or Busbars
RP	Switches
RR	Transformers
RS	Units
	Powertrain Transmission Control Systems
TA	Transmission controls
TC	Auto clutch
TD	Axle/differential
	Other
XL	Temporary numbering for future features

WARNINGS

- *Always wear safety glasses for eye protection.*
- *Use safety stands whenever a procedure requires being under a vehicle.*
- *Be sure that the* **Ignition Switch** *is always in the OFF position, unless otherwise required by the procedure.*
- *Set the parking brake when working on any vehicle. An automatic transmission should be in PARK. A manual transmission should be in NEUTRAL.*
- *Operate the engine only in a well-ventilated area to avoid danger of carbon monoxide.*
- *Keep away from moving parts, especially the fan and belts, when the engine is running.*
- *To prevent serious burns, avoid contact with hot metal parts such as the radiator, exhaust manifold, tail pipe, catalytic converter and muffler.*
- *Do not allow flame or sparks near the battery. Gases are always present in and around the battery cell. An explosion could occur.*
- *Do not smoke when working on a vehicle.*
- *To avoid injury, always remove rings, watches, loose hanging jewelry and avoid wearing loose clothing.*